10. Let X be a metric space, and let G be an open set in X. Prove that G is disjoint from a set $A \Leftrightarrow G$ is disjoint from $\bar{A}$.
11. Prove the facts about boundaries stated in the text.
12. Describe the boundary of each of the following subsets of the real line: the integers; the rationals; $[0,1]$; $(0,1)$. Do the same for each of the following subsets of the complex plane: $\{z:|z| < 1\}$; $\{z:|z| \leq 1\}$; $\{z:I(z) > 0\}$.
13. Let X be a metric space and A a subset of X. A is said to be *dense* (or *everywhere dense*) if $\bar{A} = X$. Prove that A is dense $\Leftrightarrow$ the only closed superset of A is $X \Leftrightarrow$ the only open set disjoint from A is $\emptyset \Leftrightarrow A$ intersects every non-empty open set $\Leftrightarrow A$ intersects every open sphere.

12. CONVERGENCE, COMPLETENESS, AND BAIRE'S THEOREM

As we emphasized in the introduction to this chapter, one of our main aims in considering metric spaces is to study convergent sequences in a context more general than that of classical analysis. The fruits of this study are many, and among them is the added insight gained into ordinary convergence as it is used in analysis.

Let X be a metric space with metric d, and let

$$\{x_n\} = \{x_1, x_2, \ldots, x_n, \ldots\}$$

be a sequence of points in X. We say that $\{x_n\}$ is *convergent* if there exists a point x in X such that either

(1) for each $\epsilon > 0$, there exists a positive integer n_0 such that $n \geq n_0 \Rightarrow d(x_n,x) < \epsilon$; or equivalently,

(2) for each open sphere $S_\epsilon(x)$ centered on x, there exists a positive integer n_0 such that x_n is in $S_\epsilon(x)$ for all $n \geq n_0$.

The reader should observe that the first condition is a direct generalization of convergence for sequences of numbers as defined in the introduction, and that the second can be thought of as saying that each open sphere centered on x contains all points of the sequence from some place on. If we rely on our knowledge of what is meant by a convergent sequence of real numbers, the statement that $\{x_n\}$ is convergent can equally well be defined as follows: there exists a point x in X such that $d(x_n,x) \to 0$. We usually symbolize this by writing

$$x_n \to x,$$

and we express it verbally by saying that x_n *approaches* x, or that x_n *converges to* x. It is easily seen from condition (2) and Problem 10-1 that the point x in this discussion is unique, that is, that $x_n \to y$ with $y \neq x$

is impossible. The point x is called the *limit* of the sequence $\{x_n\}$, and we sometimes write $x_n \to x$ in the form

$$\lim x_n = x.$$

The statements $x_n \to x$ and $\lim x_n = x$

mean exactly the same thing, namely, that $\{x_n\}$ is a convergent sequence with limit x.

Every convergent sequence $\{x_n\}$ has the following property: for each $\epsilon > 0$, there exists a positive integer n_0 such that $m, n \geq n_0 \Rightarrow d(x_m,x_n) < \epsilon$. For if $x_n \to x$, then there exists a positive integer n_0 such that $n \geq n_0 \Rightarrow d(x_n,x) < \epsilon/2$, and from this we see that

$$m,n \geq n_0 \Rightarrow d(x_m,x_n) \leq d(x_m,x) + d(x,x_n) < \epsilon/2 + \epsilon/2 = \epsilon.$$

A sequence with this property is called a *Cauchy sequence*, and we have just shown that every convergent sequence is a Cauchy sequence. Loosely speaking, this amounts to the statement that if the terms of a sequence approach a limit, then they get close to one another. It is of basic importance to understand that the converse of this need not be true, that is, that a Cauchy sequence is not necessarily convergent. As an example, consider the subspace $X = (0,1]$ of the real line. The sequence defined by $x_n = 1/n$ is easily seen to be a Cauchy sequence in this space, but it is not convergent, since the point 0 (which it wants to converge to) is not a point of the space. The difficulty which arises in this example stems from the fact that the notion of a convergent sequence is not intrinsic to the sequence itself, but also depends on the structure of the space in which it lies. A convergent sequence is not convergent "on its own"; it must converge to some point in the space. Some writers emphasize the distinction between convergent sequences and Cauchy sequences by calling the latter "intrinsically convergent" sequences.

A *complete metric space* is a metric space in which every Cauchy sequence is convergent. In rough terms, a metric space is complete if every sequence in it which tries to converge is successful, in the sense that it finds a point in the space to converge to. The space $(0,1]$ mentioned above is not complete, but it evidently can be made so by adjoining the point 0 to it to form the slightly larger space $[0,1]$. As a matter of fact, any metric space, if it isn't already complete, can be made so by suitably adjoining additional points. We outline this process in a problem at the end of Sec. 14.

It is a fundamental fact of elementary analysis that the real line is a complete metric space. The complex plane is also complete, as we see from the following argument. Let $\{z_n\}$, where $z_n = a_n + ib_n$, be a Cauchy sequence of complex numbers. Then $\{a_n\}$ and $\{b_n\}$ are them-

selves Cauchy sequences of real numbers, since

$$|a_m - a_n| \leq |z_m - z_n|$$

and

$$|b_m - b_n| \leq |z_m - z_n|.$$

By the completeness of the real line, there exist real numbers a and b such that $a_n \to a$ and $b_n \to b$. If we now put $z = a + ib$, then we see that $z_n \to z$ by means of

$$\begin{aligned} |z_n - z| &= |(a_n + ib_n) - (a + ib)| \\ &= |(a_n - a) + i(b_n - b)| \\ &\leq |a_n - a| + |b_n - b| \end{aligned}$$

and the fact that both final terms on the right approach 0. The completeness of the complex plane thus depends directly on the completeness of the real line. The metric space defined in Example 9-1 is also complete; for in this space a Cauchy sequence must be constant (i.e., it must consist of a single point repeated) from some place on, and it converges with that point as its limit.

The first three of the five metric spaces given as examples in Sec. 9 are therefore complete. What about the last two?

We ask the reader to show in Problem 5 that Example 9-4 is not complete. The problem of completing this space leads to the modern theory of Lebesgue integration, and it would carry us too far afield to pursue this matter to its natural conclusion.

On the other hand, the space $\mathcal{C}[0,1]$ defined in Example 9-5 is complete. We prove this in a more general form in Sec. 14. The completeness of this space, and of others similar to it, is one of the major focal points of topology and modern analysis.

The terms *limit* and *limit point* are often a source of confusion for people not thoroughly accustomed to them. On the real line, for instance, the constant sequence $\{1, 1, \ldots, 1, \ldots\}$ is convergent with limit 1; but the set of points of this sequence is the set consisting of the single element 1, and by Problem 11-2, the point 1 is not a limit point of this set. The essence of the matter is that a sequence of points in a set is not a subset of the set: it is a function defined on the positive integers with values in the set, and is usually specified by listing its values, as in $\{x_n\} = \{x_1, x_2, \ldots, x_n, \ldots\}$, where x_n is the value of the function at the integer n. A sequence may have a limit, but cannot have a limit point; and the set of points of a sequence may have a limit point, but cannot have a limit. The following theorem relates these concepts to one another and is a useful tool for some of our later work.

Theorem A. *If a convergent sequence in a metric space has infinitely many distinct points, then its limit is a limit point of the set of points of the sequence.*

PROOF. Let X be a metric space, and let $\{x_n\}$ be a convergent sequence in X with limit x. We assume that x is not a limit point of the set of points of the sequence, and we show that it follows from this that the sequence has only finitely many distinct points. Our assumption implies that there exists an open sphere $S_r(x)$ centered on x which contains no point of the sequence different from x. However, since x is the limit of the sequence, all x_n's from some place on must lie in $S_r(x)$, hence must coincide with x. From this we see that there are only finitely many distinct points in the sequence.

Our next theorem guarantees the completeness of many metric spaces which arise as subspaces of complete metric spaces.

Theorem B. *Let X be a complete metric space, and let Y be a subspace of X. Then Y is complete $\Leftrightarrow$ it is closed.*

PROOF. We assume first that Y is complete as a subspace of X, and we show that it is closed. Let y be a limit point of Y. For each positive integer n, $S_{1/n}(y)$ contains a point y_n in Y. It is clear that $\{y_n\}$ converges to y in X and is a Cauchy sequence in Y, and since Y is complete, y is in Y. Y is therefore closed.

We now assume that Y is closed, and we show that it is complete. Let $\{y_n\}$ be a Cauchy sequence in Y. It is also a Cauchy sequence in X, and since X is complete, $\{y_n\}$ converges to a point x in X. We show that x is in Y. If $\{y_n\}$ has only finitely many distinct points, then x is that point infinitely repeated and is thus in Y. On the other hand, if $\{y_n\}$ has infinitely many distinct points, then, by Theorem A, x is a limit point of the set of points of the sequence; it is therefore also a limit point of Y, and since Y is closed, x is in Y.

A sequence $\{A_n\}$ of subsets of a metric space is called a *decreasing* sequence if

$$A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots .$$

The following theorem gives conditions under which the intersection of such a sequence is non-empty.

Theorem C (Cantor's Intersection Theorem). *Let X be a complete metric space, and let $\{F_n\}$ be a decreasing sequence of non-empty closed subsets of X such that $d(F_n) \to 0$. Then $F = \bigcap_{n=1}^{\infty} F_n$ contains exactly one point.*

PROOF. It is first of all evident from the assumption $d(F_n) \to 0$ that F cannot contain more than one point, so it suffices to show that F is non-empty. Let x_n be a point in F_n. Since $d(F_n) \to 0$, $\{x_n\}$ is a Cauchy sequence. Since X is complete, $\{x_n\}$ has a limit x. We show that x is in F, and for this it suffices to show that x is in F_{n_0} for a fixed but arbitrary n_0. If $\{x_n\}$ has only finitely many distinct points, then x is that point

infinitely repeated, and is therefore in F_{n_0}. If $\{x_n\}$ has infinitely many distinct points, then x is a limit point of the set of points of the sequence, it is a limit point of the subset $\{x_n: n \geq n_0\}$ of the set of points of the sequence, it is a limit point of F_{n_0}, and thus (since F_{n_0} is closed) it is in F_{n_0}.

A subset A of a metric space is said to be *nowhere dense* if its closure has empty interior. It is easy to see that A is nowhere dense $\Leftrightarrow$ $\bar{A}$ does not contain any non-empty open set $\Leftrightarrow$ each non-empty open set has a non-empty open subset disjoint from $\bar{A}$ $\Leftrightarrow$ each non-empty open set has a non-empty open subset disjoint from A $\Leftrightarrow$ each non-empty open set contains an open sphere disjoint from A. If a nowhere dense set is thought of as a set which doesn't cover very much of the space, then our next theorem says that a complete metric space cannot be covered by any sequence of such sets.

Theorem D. *If $\{A_n\}$ is a sequence of nowhere dense sets in a complete metric space X, then there exists a point in X which is not in any of the A_n's.*

PROOF. For the duration of this proof, we abandon our usual notations for open spheres and closed spheres. Since X is open and A_1 is nowhere dense, there is an open sphere S_1 of radius less than 1 which is disjoint from A_1. Let F_1 be the concentric closed sphere whose radius is one-half that of S_1, and consider its interior. Since A_2 is nowhere dense, $\text{Int}(F_1)$ contains an open sphere S_2 of radius less than $\frac{1}{2}$ which is disjoint from A_2. Let F_2 be the concentric closed sphere whose radius is one-half that of S_2, and consider its interior. Since A_3 is nowhere dense, $\text{Int}(F_2)$ contains an open sphere S_3 of radius less than $\frac{1}{4}$ which is disjoint from A_3. Let F_3 be the concentric closed sphere whose radius is one-half that of S_3. Continuing in this way, we get a decreasing sequence $\{F_n\}$ of non-empty closed subsets of X such that $d(F_n) \to 0$. Since X is complete, Theorem C guarantees that there exists a point x in X which is in all the F_n's. This point is clearly in all the S_n's, and therefore (since S_n is disjoint from A_n) it is not in any of the A_n's.

For our purposes, the following equivalent form of Theorem D is often more convenient.

Theorem E. *If a complete metric space is the union of a sequence of its subsets, then the closure of at least one set in the sequence must have non-empty interior.*

Theorems D and E are really one theorem expressed in two different ways. We refer to both (or either) as *Baire's theorem.* This theorem is admittedly rather technical in nature, and the reader can hardly be expected to appreciate its significance at the present stage of our work.

He will find, however, that a need for it crops up from time to time, and when this need arises, Baire's theorem is an indispensable tool.[1]

Problems

1. Let X be a metric space. If $\{x_n\}$ and $\{y_n\}$ are sequences in X such that $x_n \to x$ and $y_n \to y$, show that $d(x_n,y_n) \to d(x,y)$.
2. Show that a Cauchy sequence is convergent $\Leftrightarrow$ it has a convergent subsequence.
3. If $X = X_1 \times X_2 \times \cdots \times X_n$ is the product in Problem 9-4, and if each of the coordinate spaces $X_1, X_2, \ldots, X_n$ is complete, show that X is complete with respect to each of the metrics d and $\bar{d}$ defined in that problem.
4. Let X be any non-empty set. By Problem 9-5, the set of all bounded real functions defined on X is a metric space with respect to the metric induced by the norm defined in that problem. Show that this metric space is complete. (*Hint:* if $\{f_n\}$ is a Cauchy sequence, then $\{f_n(x)\}$ is a Cauchy sequence of real numbers for each point x in X.)
5. In Example 9-4, show that the following functions f_n defined on [0,1] form a Cauchy sequence in this space which is not convergent: $f_n(x) = 1$ if $0 \leq x \leq \frac{1}{2}$, $f_n(x) = -2^n(x - \frac{1}{2}) + 1$ if $\frac{1}{2} \leq x \leq \frac{1}{2} + (\frac{1}{2})^n$, and $f_n(x) = 0$ if $\frac{1}{2} + (\frac{1}{2})^n \leq x \leq 1$.
6. Give an example to show that the set F in Cantor's intersection theorem may be empty if the hypothesis $d(F_n) \to 0$ is dropped.
7. Show that a closed set is nowhere dense $\Leftrightarrow$ its complement is everywhere dense.
8. Show that the Cantor set is nowhere dense.

13. CONTINUOUS MAPPINGS

In the previous section we extended the idea of convergence to the context of a general metric space. We now do the same for continuity.

Let X and Y be metric spaces with metrics d_1 and d_2, and let f be a mapping of X into Y. f is said to be *continuous at a point* x_0 *in* X if either

[1] There is some rather undescriptive terminology which is often used in connection with Baire's theorem. We shall not make use of it ourselves, but the reader ought to be acquainted with it. A subset of a metric space is called a set of the *first category* if it can be represented as the union of a sequence of nowhere dense sets, and a set of the *second category* if it is not a set of the first category. Baire's theorem—sometimes called the *Baire category theorem*—can now be expressed as follows: any complete metric space (considered as a subset of itself) is a set of the second category.

of the following equivalent conditions is satisfied:

(1) for each $\epsilon > 0$ there exists $\delta > 0$ such that $d_1(x,x_0) < \delta \Rightarrow d_2(f(x),f(x_0)) < \epsilon$;

(2) for each open sphere $S_\epsilon(f(x_0))$ centered on $f(x_0)$ there exists an open sphere $S_\delta(x_0)$ centered on x_0 such that $f(S_\delta(x_0)) \subseteq S_\epsilon(f(x_0))$.

The reader will notice that the first condition generalizes the elementary definition given in the introduction to this chapter, and that the second translates the first into the language of open spheres.

Our first theorem expresses continuity at a point in terms of sequences which converge to the point.

Theorem A. *Let X and Y be metric spaces and f a mapping of X into Y. Then f is continuous at x_0 if and only if $x_n \to x_0 \Rightarrow f(x_n) \to f(x_0)$.*

PROOF. We first assume that f is continuous at x_0. If $\{x_n\}$ is a sequence in X such that $x_n \to x_0$, we must show that $f(x_n) \to f(x_0)$. Let $S_\epsilon(f(x_0))$ be an open sphere centered on $f(x_0)$. By our assumption, there exists an open sphere $S_\delta(x_0)$ centered on x_0 such that $f(S_\delta(x_0)) \subseteq S_\epsilon(f(x_0))$. Since $x_n \to x_0$, all x_n's from some place on lie in $S_\delta(x_0)$. Since $f(S_\delta(x_0)) \subseteq S_\epsilon(f(x_0))$, all $f(x_n)$'s from some place on lie in $S_\epsilon(f(x_0))$. We see from this that $f(x_n) \to f(x_0)$.

To prove the other half of our theorem, we assume that f is not continuous at x_0, and we show that $x_n \to x_0$ does not imply $f(x_n) \to f(x_0)$. By this assumption, there exists an open sphere $S_\epsilon(f(x_0))$ with the property that the image under f of each open sphere centered on x_0 is not contained in it. Consider the sequence of open spheres $S_1(x_0)$, $S_{1/2}(x_0)$, . . . , $S_{1/n}(x_0)$, ... Form a sequence $\{x_n\}$ such that $x_n \in S_{1/n}(x_0)$ and $f(x_n) \notin S_\epsilon(f(x_0))$. It is clear that x_n converges to x_0 and that $f(x_n)$ does not converge to $f(x_0)$.

A mapping of one metric space into another is said to be *continuous* if it is continuous at each point in its domain. The following theorem is an immediate consequence of Theorem A and this definition.

Theorem B. *Let X and Y be metric spaces and f a mapping of X into Y. Then f is continuous if and only if $x_n \to x \Rightarrow f(x_n) \to f(x)$.*

This result shows that continuous mappings of one metric space into another are precisely those which send convergent sequences into convergent sequences, or, in other words, which *preserve convergence.* Our next theorem characterizes continuous mappings in terms of open sets.

Theorem C. *Let X and Y be metric spaces and f a mapping of X into Y. Then f is continuous $\Leftrightarrow f^{-1}(G)$ is open in X whenever G is open in Y.*

PROOF. We first assume that f is continuous. If G is an open set in Y, we must show that $f^{-1}(G)$ is open in X. $f^{-1}(G)$ is open if it is empty, so

we may assume that it is non-empty. Let x be a point in $f^{-1}(G)$. Then $f(x)$ is in G, and since G is open, there exists an open sphere $S_\epsilon(f(x))$ centered on $f(x)$ and contained in G. By the definition of continuity, there exists an open sphere $S_\delta(x)$ such that $f(S_\delta(x)) \subseteq S_\epsilon(f(x))$. Since $S_\epsilon(f(x)) \subseteq G$, we also have $f(S_\delta(x)) \subseteq G$, and from this we see that $S_\delta(x) \subseteq f^{-1}(G)$. $S_\delta(x)$ is therefore an open sphere centered on x and contained in $f^{-1}(G)$, so $f^{-1}(G)$ is open.

We now assume that $f^{-1}(G)$ is open whenever G is, and we show that f is continuous. We show that f is continuous at an arbitrary point x in X. Let $S_\epsilon(f(x))$ be an open sphere centered on $f(x)$. This open sphere is an open set, so its inverse image is an open set which contains x. By this, there exists an open sphere $S_\delta(x)$ which is contained in this inverse image. It is clear that $f(S_\delta(x))$ is contained in $S_\epsilon(f(x))$, so f is continuous at x. Finally, since x was taken to be an arbitrary point in X, f is continuous.

The fact just established—that continuous mappings are precisely those which pull open sets back to open sets—will be of great importance for all our work from Chap. 3 on.

We now come to the useful concept of *uniform* continuity. In order to explain what this is, we examine the definition of continuity expressed in condition (1) at the beginning of this section. Let X and Y be metric spaces with metrics d_1 and d_2, and let f be a mapping of X into Y. We assume that f is continuous, that is, that for each point x_0 in X the following is true: given $\epsilon > 0$, a number $\delta > 0$ can be found such that $d_1(x,x_0) < \delta \Rightarrow d_2(f(x),f(x_0)) < \epsilon$. The reader is no doubt familiar with the idea that if x_0 is held fixed and ϵ is made smaller, then, in general, δ has to be made correspondingly smaller. Thus, in the case of the real function f defined by $f(x) = 2x$, δ can always be chosen as any positive number $\leq \epsilon/2$, and no larger δ will do. In general, therefore, δ depends on ϵ. Let us return to our examination of the definition. It says that for our given ϵ, a δ can be found which "works" in the above sense at the particular point x_0 under consideration. But if we hold ϵ fixed and move to another point x_0, then it may happen that this δ no longer works; that is, it may be necessary to take a smaller δ to satisfy the requirement of the definition. We see in this way that δ may well depend, in general, not only on ϵ but also on x_0. Uniform continuity is essentially continuity plus the added condition that for each ϵ we can find a δ which works *uniformly over the entire space* X, in the sense that it does not depend on x_0. The formal definition is as follows. If X and Y are metric spaces with metrics d_1 and d_2, then a mapping f of X into Y is said to be *uniformly continuous* if for each $\epsilon > 0$ there exists $\delta > 0$ such that $d_1(x,x') < \delta \Rightarrow d_2(f(x),f(x')) < \epsilon$. It is clear that any uni-

formly continuous mapping is automatically continuous. The reader will observe that the above real function f defined on the entire real line R by $f(x) = 2x$ is uniformly continuous. On the other hand, the function g defined on R by $g(x) = x^2$ is continuous but not uniformly continuous. Similarly, the continuous function h defined on (0,1) by $h(x) = 1/x$ is not uniformly continuous.

Uniformly continuous mappings—as opposed to those which are merely continuous—are of particular significance in analysis. The following theorem expresses a property of these mappings which is often useful.

Theorem D. *Let X be a metric space, let Y be a complete metric space, and let A be a dense subspace of X. If f is a uniformly continuous mapping of A into Y, then f can be extended uniquely to a uniformly continuous mapping g of X into Y.*

PROOF. Let d_1 and d_2 be the metrics on X and Y. If $A = X$, the conclusion is obvious. We therefore assume that $A \neq X$. We begin by showing how to define the mapping g. If x is a point in A, we define $g(x)$ to be $f(x)$. Now let x be a point in $X - A$. Since A is dense, x is the limit of a convergent sequence $\{a_n\}$ in A. Since $\{a_n\}$ is a Cauchy sequence and f is uniformly continuous, $\{f(a_n)\}$ is a Cauchy sequence in Y (see Problem 8). Since Y is complete, there exists a point in Y—we call this point $g(x)$—such that $f(a_n) \to g(x)$. We must make sure that $g(x)$ depends only on x, and not on the sequence $\{a_n\}$. Let $\{b_n\}$ be another sequence in A such that $b_n \to x$. Then $d_1(a_n,b_n) \to 0$, and by the uniform continuity of f, $d_2(f(a_n),f(b_n)) \to 0$. It readily follows from this that $f(b_n) \to g(x)$.

We next show that g is uniformly continuous. Let $\epsilon > 0$ be given, and use the uniform continuity of f to find $\delta > 0$ such that for a and a' in A we have $d_1(a,a') < \delta \Rightarrow d_2(f(a),f(a')) < \epsilon$. Let x and x' be any points in X such that $d_1(x,x') < \delta$. It suffices to show that $d_2(g(x), g(x')) \leq \epsilon$. Let $\{a_n\}$ and $\{a'_n\}$ be sequences in A such that $a_n \to x$ and $a'_n \to x'$. By the triangle inequality, we see that $d_1(a_n,a'_n) \leq d_1(a_n,x) + d_1(x,x') + d_1(x',a'_n)$. This inequality, together with the facts that $d_1(a_n,x) \to 0$, $d_1(x,x') < \delta$, and $d_1(x',a'_n) \to 0$, implies that $d_1(a_n,a'_n) < \delta$ for all sufficiently large n. It now follows that $d_2(f(a_n),f(a'_n)) < \epsilon$ for all sufficiently large n. By Problem 12-1,

$$d_2(g(x),g(x')) = \lim d_2(f(a_n),f(a'_n)),$$

and from this and the previous statement we see that $d_2(g(x),g(x')) \leq \epsilon$.

All that remains is to show that g is unique, and this is easily proved by means of Problem 3 below.

There is an important type of uniformly continuous mapping which often arises in practice. If X and Y are metric spaces with metrics d_1 and d_2, a mapping f of X onto Y is called an *isometry* (or an *isometric mapping*) if $d_1(x,x') = d_2(f(x),f(x'))$ for all points x and x' in X; and if such a mapping exists, we say that X is *isometric* to Y. It is clear that an isometry is necessarily one-to-one. If X is isometric to Y, then the points of these spaces can be put into one-to-one correspondence in such a way that the distances between pairs of corresponding points are the same. The spaces therefore differ only in the nature of their points, and this is often unimportant. We usually consider isometric spaces to be identical with one another. It is often convenient to be able to use this terminology in the case of mappings which are not necessarily onto. If f is a mapping of X into Y which preserves distances in the above sense, then we call f an isometry of X *into* Y, or an isometry of X *onto* the subspace $f(X)$ of Y. In this situation, we often say that Y contains an *isometric image* of X, namely, the subspace $f(X)$.

Problems

1. Let X and Y be metric spaces and f a mapping of X into Y. If f is a constant mapping, show that f is continuous. Use this to show that a continuous mapping need not have the property that the image of every open set is open.
2. Let X be a metric space with metric d, and let x_0 be a fixed point in X. Show that the real function f_{x_0} defined on X by $f_{x_0}(x) = d(x,x_0)$ is continuous. Is it uniformly continuous?
3. Let X and Y be metric spaces and A a non-empty subset of X. If f and g are continuous mappings of X into Y such that $f(x) = g(x)$ for every x in A, show that $f(x) = g(x)$ for every x in $\bar{A}$.
4. Let X and Y be metric spaces and f a mapping of X into Y. Show that f is continuous $\Leftrightarrow f^{-1}(F)$ is closed in X whenever F is closed in $Y \Leftrightarrow f(\bar{A}) \subseteq \overline{f(A)}$ for every subset A of X.
5. Show that any mapping of the metric space defined in Example 9-1 into any other metric space is continuous.
6. Consider the real function f defined on the real line R by $f(x) = x^2$. If b is a given positive real number, show that the restriction of f to the closed interval $[0,b]$ is uniformly continuous by starting with an $\epsilon > 0$ and exhibiting a $\delta > 0$ which satisfies the requirement of the definition.
7. Determine which of the following functions are uniformly continuous on the open unit interval $(0,1)$: $1/(1-x)$; $1/(2-x)$; $\sin x$; $\sin(1/x)$; $x^{1/2}$; x^3. Which are uniformly continuous on the open interval $(0, +\infty)$?

8. In the proof of Theorem D we used the following fact: the image of a Cauchy sequence under a uniformly continuous mapping is again a Cauchy sequence. Give the details of the proof.
9. Let f be a continuous real function defined on R which satisfies the functional equation $f(x + y) = f(x) + f(y)$. Show that this function must have the form $f(x) = mx$ for some real number m. (*Hint:* the subspace of rational numbers is dense in the metric space R.)

14. SPACES OF CONTINUOUS FUNCTIONS

In Example 9-5 we gave a brief description of the metric space $\mathcal{C}[0,1]$. The reader will recall that the points of this space are the bounded continuous real functions defined on the closed unit interval $[0,1]$ and that its metric is defined by $d(f,g) = \sup |f(x) - g(x)|$. We have two aims in this section: to generalize this very important example by considering functions defined on an arbitrary metric space, and to place all function spaces of this type in their proper context by giving the details of the structural pattern (discussed briefly in Sec. 9) which they all have in common with one another. We begin with the second, and define the algebraic systems which are relevant to our present interests.

Let L be a non-empty set, and assume that each pair of elements x and y in L can be combined by a process called *addition* to yield an element z in L denoted by $z = x + y$. Assume also that this operation of addition satisfies the following conditions:

(1) $x + y = y + x$;
(2) $x + (y + z) = (x + y) + z$;
(3) there exists in L a unique element, denoted by 0 and called the *zero element*, or the *origin*, such that $x + 0 = x$ for every x;
(4) to each element x in L there corresponds a unique element in L, denoted by $-x$ and called the *negative* of x, such that $x + (-x) = 0$.

We adopt the device of referring to the system of real numbers or to the system of complex numbers as the *scalars*. We now assume that each scalar α and each element x in L can be combined by a process called *scalar multiplication* to yield an element y in L denoted by $y = \alpha x$ in such a way that

(5) $\alpha(x + y) = \alpha x + \alpha y$;
(6) $(\alpha + \beta)x = \alpha x + \beta x$;
(7) $(\alpha\beta)x = \alpha(\beta x)$;
(8) $1 \cdot x = x$.